

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 11

Status: active BA2 execution plan after BA2-T3 cross-corpus regression and reduction.

Supersedes: Revision 10 only for forward execution state; R1-R10 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- **BAReferent** and **BAProposition**: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. BA2 objective

BA2 defines the smallest stable methodology-neutral proposition/relation/action semantics over the BA1 identities while preserving:

```
semantic identity family
    !=
semantic operator / participation role / modifier
    !=
lexical label / authoring synonym
```

Base Analysis must support mechanical selection and projection without forcing consumers to reinterpret raw prose or import threat-method categories.

3. BA2-T1 - proposition structural lower bound

Status: COMPLETED / PROVISIONAL CANDIDATE.

Pure binary SPO is rejected as the universal structure. Explicit role-bound n-ary participation remains the lower bound.

4. BA2-T2 - vocabulary pressure test

Status: COMPLETED / PROVISIONAL PASS WITH VOCABULARY-ARCHITECTURE REFINEMENT.

T2 established stable semantic operator keys, operator-scoped role contracts, explicit polarity as a strong candidate, normalized local modifiers with a promotion rule and classification-as-proposition as a strong candidate.

5. BA2-T3 - cross-corpus regression

Status: COMPLETED / PROVISIONAL PASS WITH REDUCTION.

T3 replayed the T2 candidate across facial-access and order-fulfillment and produced BA2_SEMANTIC_VOCABULARY_CANDIDATE

Main reductions

operator-family facet	REMOVED FROM NORMATIVE CORE
seed operators	14 -> 13
ownOrManage	MERGED INTO assignResponsibility
missing operator	NONE FORCED
input role under produce	ADDED / REQUIRED BY CORPUS
responsibilityKind role	ADDED / REQUIRED BY MERGE
polarity	RETAINED
modifier set	REDUCED TO condition + temporalScope

completion/atomicity/	
concurrency/idempotency	PROMOTED TO constrain
classification-as-proposition	RETAINED
universal semantic-kind enum	NOT REQUIRED
bounded STRIDE consumer	PASS
BA1 reopen	NOT TRIGGERED

Reduced operator seed

transfer
produce
create
observe
transition
correlate
reference
dependOn
consumeService
realize
assignResponsibility
constrain
classify

This is still candidate until closure review.

6. BA2-T4 - relation/action vocabulary closure review

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA2-T4 must review the regressed R2 candidate without expanding vocabulary unless a concrete closure counterexample appears.

It must decide:

1. whether the thirteen-key operator seed is sufficient and minimal for current thesis scope;
2. whether the operator-role/cardinality matrix is coherent and does not hide method-specific assumptions;
3. whether `ownOrManage -> assignResponsibility + responsibilityKind + polarity` preserves the needed responsibility boundary;
4. whether explicit polarity plus only local `condition` and `temporalScope` modifiers is sufficient;
5. whether `atomicity/concurrency/idempotency/completion` and similar governed rules belong in `constrain` rather than modifiers;
6. whether `classification-as-proposition` plus a controlled semantic-kind registry contract is sufficient without a fixed universal taxonomy;
7. whether semantic-key/lexical-label separation is clean enough to defer labels/synonyms to BA5;
8. whether any remaining issue actually belongs to BA3 rather than BA2;
9. whether any concrete counterexample triggers BA1 reopen.

BA2-T4 exit condition

Close BA2 only if no material relation/action/role/modifier counterexample remains and all unresolved items can be allocated honestly to BA3/BA5/BA6 without weakening mechanical analysis consumption.

If a concrete counterexample appears, reopen only the smallest affected BA2 candidate responsibility. Do not broaden into general systems modeling.

7. BA3 - derivation, provenance and authority

Only after BA2 closes, BA3 must close:

- document-to-BAReferent/BAProposition derivation rules;
- source locators;
- grounded, derived and diagnostic/unresolved materialization;
- identity/equivalence/lifecycle rules;
- acceptance/rejection/stale semantics;

- source-to-analysis and analysis-to-source traceability.

Governed documentation remains project authority.

8. BA4 - projections

Define derived human and method projections from the same accepted BA semantics. No view becomes source authority.

9. BA5 - lexical vocabulary and optional assistance

Only after semantic operator/role contracts are closed, define canonical display labels, authoring synonyms and optional assistance. NLP/LLM proposals remain candidate-producing support and cannot establish project semantics silently.

10. BA6 - Base Analysis closure and regression

Regress the complete Base Analysis design against closed corpora and at least one structurally different holdout. BA6 closes the complete Base Analysis design.

11. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The earlier STRIDE transfer probe remains only a bounded consumer probe.

12. Next authorized microstep

After the BA2-T3 package is reviewed, committed and remotely verified, execute only:

BA2-T4 - relation/action vocabulary closure review over the regressed R2 candidate.

Do not start BA3, formal STRIDE overlay design, Common Finding schema or implementation work.